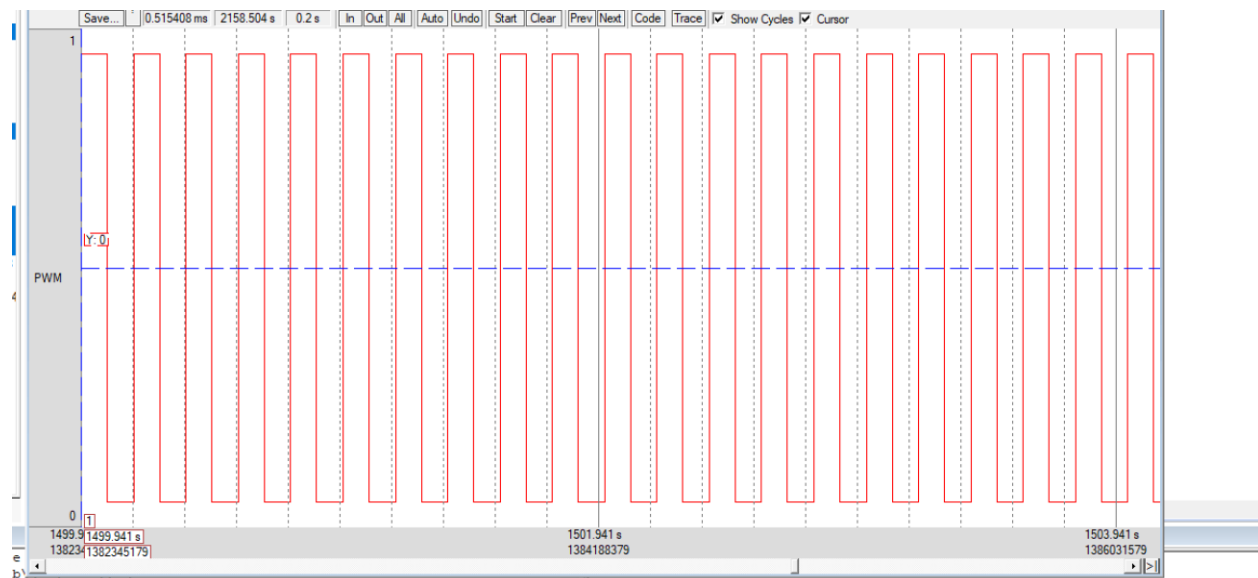


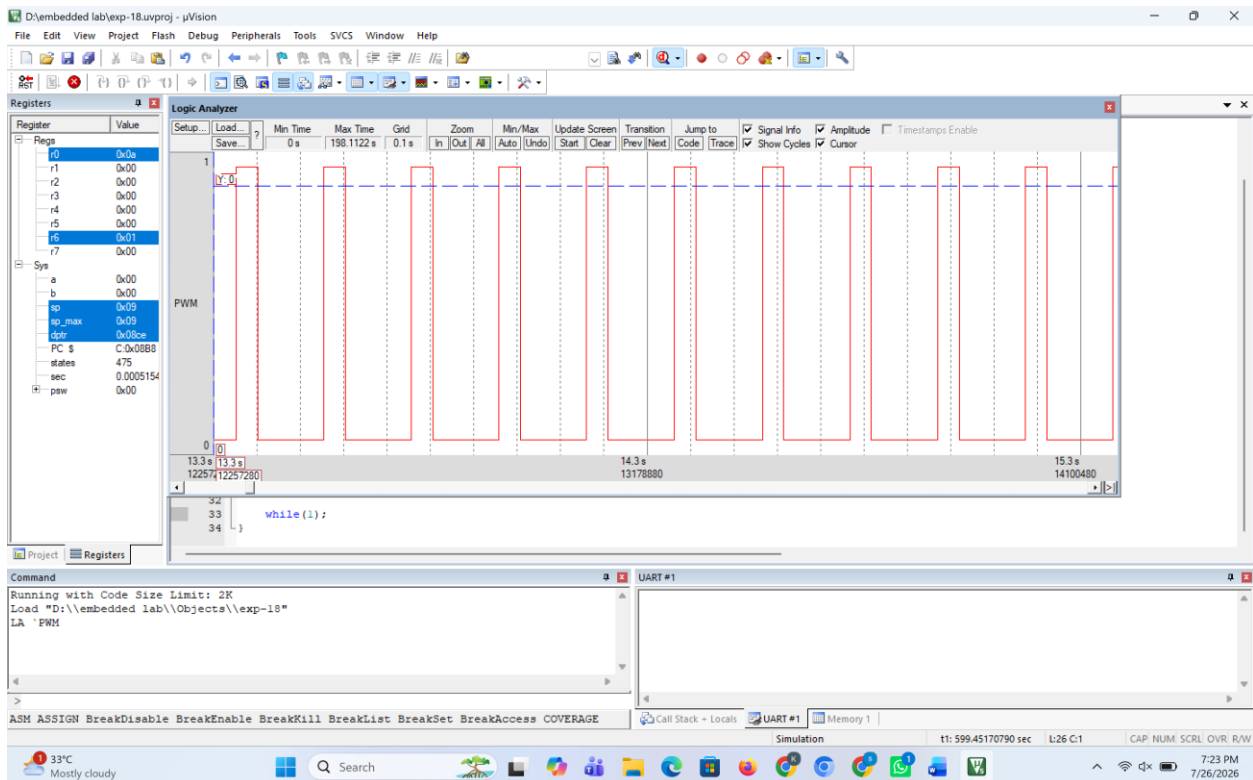
```
EXP-19.C*
1  #include <reg51.h>
2  sbit PWM = P1^0;
3  unsigned char count = 0;
4  unsigned char duty = 100;
5  void timer0_ISR(void) interrupt 1
6  {
7      TH0 = 0xFC;
8      TL0 = 0x66;
9
10     count++;
11
12     if(count < duty)
13         PWM = 1;
14     else
15         PWM = 0;
16
17     if(count >= 200)
18         count = 0;
19 }
20 void main()
21 {
22     TMOD = 0x01;
23     TH0 = 0xFC;
24     TL0 = 0x66;
25     IE = 0x82;
26     TR0 = 1;
27     while(1);
28 }
```

50% Duty cycle:



```
EXP-19.C*
1  #include <reg51.h>
2  sbit PWM = P1^0;
3  unsigned char count = 0;
4  unsigned char duty = 50;
5  void timer0_ISR(void) interrupt 1
6  {
7      TH0 = 0xFC;
8      TL0 = 0x66;
9
10     count++;
11
12     if(count < duty)
13         PWM = 1;
14     else
15         PWM = 0;
16
17     if(count >= 200)
18         count = 0;
19 }
20 void main()
21 {
22     TMOD = 0x01;
23     TH0 = 0xFC;
24     TL0 = 0x66;
25     IE = 0x82;
26     TR0 = 1;
27     while(1);
28 }
```

25% Duty cycle:



```
EXP-19.C*
1  #include <reg51.h>
2  sbit PWM = P1^0;
3  unsigned char count = 0;
4  unsigned char duty = 150;
5  void timer0_ISR(void) interrupt 1
6  {
7      TH0 = 0xFC;
8      TLO = 0x66;
9
10     count++;
11
12     if(count < duty)
13         PWM = 1;
14     else
15         PWM = 0;
16
17     if(count >= 200)
18         count = 0;
19 }
20 void main()
21 {
22     TMOD = 0x01;
23     TH0 = 0xFC;
24     TLO = 0x66;
25     IE = 0x82;
26     TR0 = 1;
27     while(1);
28 }
```

75% Duty cycle:

